PROJECT PROPOSAL:

- **Project description**: Rhythm is a program devoted to aiding runners and cyclists in the gym (treadmills or steady bikes) in syncing the tempo of the music they listen to, to the cadence of their feet movement, all while providing an enhanced POV experience. This has the purpose of helping them maintain a constant rhythm in their sport. It will consist of two modes:

Note: to simulate the pace of the steps of the user, I will do it through the tempo of pressing the spacebar.

Backend

- Competitive mode: the user manually sets a desired pace (for example: time running and desired distance run), and rhythm will alter the tempo of every song in your playlist, so that if you follow the rhythm with your feet, you'll make the objective.

-> The user will be able to increase or decrease the pace through a virtual slider through opencv.

- Follow mode: you input a rhythm through the space bar you feel comfortable running at, and then rhythm detects through an "on the fly" clicks per second algorithm the tempo of your spacebar clicks, and adapts the tempo of the songs to that rhythm.

Graphics

- Graphics: the graphics will consist of a POV 3D simulation of the real-life experience of scenery passing by as you run / cycle.

Note: if I manage to finish these two modes and have some time to spare, I will add a multiplayer so that different users can race each other online.

- **Similar projects**:

Policarpo Baquera's Vaporscape and N-back Racer Demo are similar when it comes to the graphics of the project.

When it comes to the backend, my project will be different because of the altering of the tempo of the songs, but similar when it comes to the opencv interaction with the user (whether it be increasing pace with finger movements, or other)

- **Structural plan**:

I will be using a Button() class for all the buttons in graphics, a Song() class to organize each song and its properties, a Counter() class to keep track of distance run, time left, and pace.

I organize the program in the following manner:

- I will have a separate file with all the various modes and low-level graphics functions, such as loadGif()
- A separate file named frontend.py that will handle the high-level graphics of the project
- An objects.py file where I will store all the classes I'll be using
- A backend.py file where I'll have both opencv and audio handling
- I'll store all the .wav audio files in a folder called 'songs' and all the required images in a folder called 'images'

- **Algorithmic plan**:

Firstly, I will properly condense and organize my code through:

- Using OOP for organizing the code, such as: button objects, song objects, etc
- Defining a different mode for the graphics of each of my three screens:
  - Home (Start mode)
  - Competitive mode
  - Following mode

In the follow mode, I will use a "clicks per second" algorithm to determine the number of spacebar presses per second on a running basis.

- **Timeline plan**:

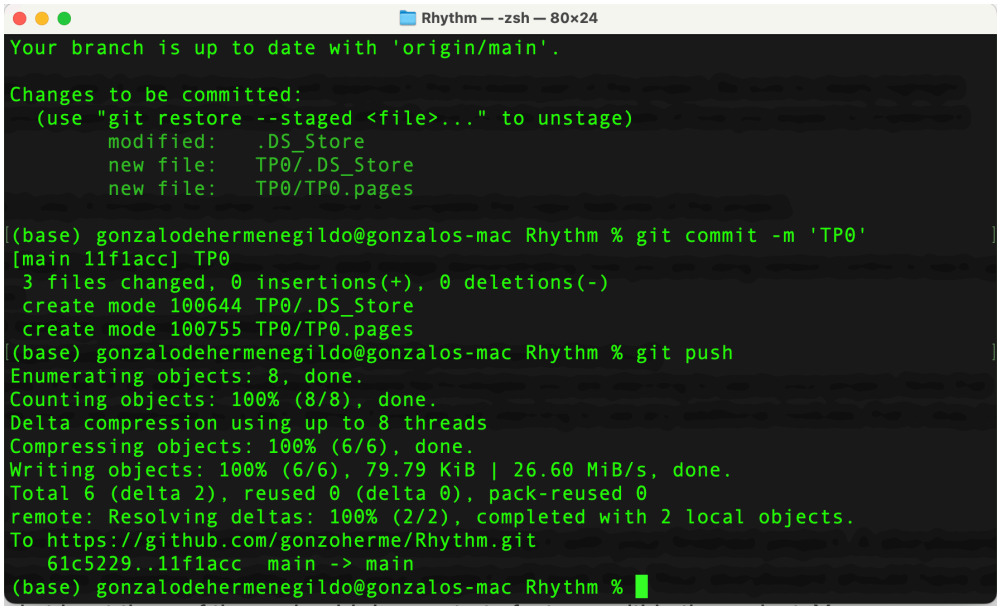
We have around 4 weeks left:

Week 1: I have already coded the speeding up and down of the songs, so I'll devote this week to making the audio interact with cmu_112_graphics. I will also get some simple opencv features to interact with the graphics.

Week 2 : I will code and incorporate the clicks per second algorithm to the follow mode. I'll also code the input boxes for competitive mode shown in the storyboard. I'll get started with the 3D graphics.

Week 3: I will code the graphics part of the POV landscape moving while the user runs. Backup plan: if I don't end up having time for this, I will play a video POV of a jogger or cyclist, and set the speed of the video in accordance to the pace of the user.

- **Version control plan**: I am using git as my version control plan, both locally and online (through gitHub)



gonzoherme Delete .DS_Store `a3bfe48` now **8** commits

TP0	TP0	2 minutes ago
__pycache__	OOP implementation 2	yesterday
images	OOP implementation 2	yesterday
mp_env	Necessary	2 days ago
songs	Update	2 days ago
README.org	Update too	2 days ago
backend.py	OOP implementation 2	yesterday
cmu_112_graphics.py	First Commit	6 days ago
frontend.py	OOP implementation 2	yesterday
module_manager.py	Necessary	2 days ago
opencv.py	Update	2 days ago

No description, website, or topics provided.

Readme
0 stars
1 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

- **Module list**: opencv, librosa